INCH-POUND
MIL-DTL-24640/4E
8 November 2011
SUPERSEDING
MIL-DTL-24640/4D
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, -20 TO +105 °C, 600 VOLTS, LIGHTWEIGHT, MULTI-PAIR, 20 AWG, TYPES TTX AND TTXWB

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24640.

Construction, Non-Watertight (TTX) and Watertight (TTXWB)

First	-	19-strand, tin-coated copper conductor, 20 AWG. Watertight constructions shall be constructed with 7 strands.
Second	-	Component wires shall be SAE-AS81044/12, except identification shall be in accordance with the codes and methods of MIL-DTL-24640. Testing shall be in accordance with the component wire ratings and requirements listed on this specification sheet. Conformance testing shall be in accordance with SAE-AS81044 prior to cabling.
Third	-	Two component wires, one black and one white, shall be twisted together to form the required number of pairs. The pitch shall be 1±¼ inch. Identification codes applied using Method 6.
Fourth	-	The required number of pairs shall be cabled together.
Fifth	-	Optional binder tape.
Sixth	-	Cross-linked polyolefin jacket (see table I for thickness). NOTE: Manufacturer's identification tape or thread is required (see 3.7.3).

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TABLE I. Construction details.

PIN M24640/4	Type number	Number of pairs	Jacket thickness (inch) (min.)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max.)
				(min.)	(max.)	
-01UN	TTX-3	3	0.038	0.296	0.320	62.1
-02UN	TTX-15	15	0.044	0.549	0.591	221.0
-03UN	TTXWB-3	3	0.038	0.310	0.330	69.5
-04UN	TTXWB-15	15	0.045	0.550	0.591	243.9

REQUIREMENTS:

Qualification required.

Construction details: See table I.

Component wire: The component wire shall be in accordance with SAE-AS81044 and shall be tested in accordance with the following wire ratings and additional requirements.

Temperature rating	150 °C (302 °F) (max.) conductor temperature
Voltage rating	600 volts (root mean square) at sea level
Accelerated aging	
Non-blocked	Oven temperature, 300±2 °C (572±3.6 °F) for 6 hours
Water-blocked	Oven temperature, 250±2 °C (482±3.6 °F) for 6 hours
Blocking	150±2 °C (302±3.6 °F)
Color	In accordance with MIL-STD-104, Class 1; white preferred
Flammability	30 seconds (max.)
	3.0 inches (max.)
	No flaming of tissue paper
Humidity resistance	5000 megohms for 1000 feet (min.) insulation resistance after humidity exposure
Dielectric test (100% test)	
Primary insulation	Impulse test - 6.0 kV (peak) or 3 kHz spark test - 1.5kV root mean square
Component wire	Impulse test - 8.0 kV (peak) or 3 kHz spark test - 5.5kV root mean square
Temperature rating	150 °C (302 °F) (max.) conductor temperature
Insulation resistance	5000 megohms for 1000 feet (min.)
Life cycle	
Non-blocked	Oven temperature, 200±2 °C (392±3.6 °F) for 168 hours
Water-blocked	Oven temperature, 200±2 °C (392±3.6 °F) for 120 hours

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Physical properties of primary insulation	
Tensile strength	2500 lb/in^2 (min.)
Elongation	150 percent (min.)
Shrinkage	
Non-blocked	0.125 inch (max.) at 300±2 °C (572±3.6 °F)
Water-blocked	0.125 inch (max.) at 250±2 °C (482±3.6 °F)
Surface resistance	500 megohms inches (min.), initial and final readings
Thermal shock	Oven temperature, 150±2 °C (302±3.6 °F)
Maximum change in measurement	0.060 inch
Wet dielectric test	2500 volts (root mean square)
Wrap test	Mandrel wrap test required

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	10.95
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Conductor to shield	Not applicable
Shield to shield	Not applicable
Insulation resistance (megohms/1000 feet, min.)	2500
Conductor continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure
Capacitance	
Mutual capacitance (at 1 MHz, picofarads/foot (pF/ft), max.)	Not applicable
Capacitance unbalance (percent, max.)	Not applicable
Characteristic impedance (at 1 MHz, ohms)	Not applicable
Watertightness (Type TTXWB only) (see table IX of MIL-DTL-24640 for limits of water leakage)	No failure
Weight	No failure

Group B:

Cold bending	No failure
Cross-link proof test (percent, max.)	

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Jacket	50
Drip (Type TTXWB only)	No failure
Tear (lb/in thickness, min.)	35
Physicals (unaged)	
Jacket	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Attenuation (at 3 MHz, dB/100 feet, max.)	Not applicable
Gas flame (1 hour)	Not applicable

Group C:

Physicals (aged), air oven	
Jacket	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (conductor insulation, cycles, Method 1 only, min.)	25
Permanence of printing (jacket, cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable sealant removability (Type TTXWB only)	No failure
Shrinkage	No failure
Surface transfer impedance	
Milliohms per meter (max.)	Not applicable
EMP response (dB, min.)	Not applicable

Group D:

Flame propagation	No failure

Qualification Inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility cable	No failure
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	

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Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	10
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Component wire	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
All types and sizes	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2009-072)

Review Activities:
Army – AR, AV, CR, CR4
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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